

PERFORMANCE TASK 3

File System Administration: Provisioning, Partitioning and Mounting Storage

IT 4112 — System Administration and Maintenance • Week 4 • Laboratory

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Date Performed:	October 1, 2026	Date Submitted:	
Duration:	2 hours 30 minutes	Total Points:	100

1 The Situation

ISLA VERDE TRADING CORP. — WORK ORDER #IVT-2026-041

Isla Verde Trading Corp. has 40 employees across Sales, Finance and Warehouse. Its single Ubuntu Server has been running on one 25 GB virtual disk since it was set up by a contractor who has since left. Nobody documented anything.

Last Tuesday the server started throwing "No space left on device" errors. The Finance team could not save month-end reports. Management approved the purchase of two new disks, and they have now been handed to you — still blank, still unpartitioned, still invisible to the operating system.

Your job as junior system administrator: bring both disks into service, give each department a properly formatted and mounted storage area, prove that files can be written to each one, then take the storage back offline cleanly at the end of your shift so the Data Center team can move the physical trays.

The two disks you have been given

Disk	Capacity	Partition Plan	File System	Intended Department
Storage 1	20 GB	1 partition — full disk	ext4	Company archive
Storage 2	30 GB	Partition 1 — 15 GB	ext4	Finance records
		Partition 2 — remaining ~15 GB	NTFS	Windows exchange folder

WHY NTFS ON A LINUX SERVER?

Finance still runs two Windows accounting workstations. Files placed on the NTFS partition can be read directly if that disk is ever moved to a Windows machine. This is a deliberate design decision, not a mistake — and one of the concept questions will ask you to defend it.

2 What You Should Be Able to Do Afterwards

By the end of this session you should be able to:

- Attach new virtual hard disks to an existing VirtualBox guest and confirm the operating system sees them.
- Read block device names correctly and explain what each part of a name such as `/dev/sdc2` means.
- Create partitions on a raw disk using `fdisk`, including a mixed layout on a single disk.
- Format partitions with the `ext4` and `NTFS` file systems and explain why you would pick one over the other.
- Create mount points, mount partitions, write data to them, and unmount them without corrupting anything.
- Document your work with evidence that another administrator could verify.

3 Before You Start

3.1 What you need

Item	Requirement
Host machine	Any machine running VirtualBox 7.x with at least 60 GB free on the physical drive
Virtual machine	Your existing Ubuntu Server guest from PT1 and PT2, powered off
Account	A user account with <code>sudo</code> privileges
Internet	Needed once, to install the <code>ntfs-3g</code> package
Software	Microsoft Word (or Google Docs / LibreOffice Writer exporting to <code>.docx</code>)

3.2 Take a snapshot first — this is not optional

You are about to run commands that permanently erase disks. A snapshot is your undo button. Take it while the VM is powered OFF.

VirtualBox Manager

```
# In the VirtualBox Manager window:
#   Right-click your VM -> Snapshots -> Take (camera icon)
#   Name it exactly: PT3-START
#   Description:    before adding storage 1 and storage 2
```

THE ONE RULE THAT MATTERS TODAY

Every destructive command in this task names a specific device. `/dev/sda` is your existing system disk. If you ever type `sda` where the instruction says `sdb` or `sdc`, you will destroy the operating system you are working on. Read the device name out loud before you press Enter.

3.3 How to spend your two and a half hours

Part	Activity	Time
Setup	Snapshot, power off, read through the whole task once	10 min

Part	Activity	Time
Part A	Attach the two virtual disks in VirtualBox	15 min
Part B	Boot and identify the new disks	10 min
Part C	Partition and format Storage 1	20 min
Part D	Partition and format Storage 2 (two partitions, two file systems)	35 min
Part E	Create mount points and mount all three partitions	20 min
Part F	Write one text file to each partition and verify	15 min
Part G	Unmount everything and confirm	10 min
Part H	Answer the concept questions	15 min
	TOTAL	2 h 30 min

4 How Your Evidence Must Look

Every part of this task asks for one or more screenshots. Fourteen screenshots are required in total. They are the only proof that you actually did the work, so they have to be readable and they have to be yours.

Rules for screenshots

- Capture the whole terminal window or the whole VirtualBox dialog — not a cropped fragment of one line.
- The command AND its output must both be visible in the same shot.
- Before your first screenshot, run `whoami` once so your username appears in the shell prompt. That prompt must be visible in every later screenshot.
- Do not retype output into the document. A pasted block of text is not a screenshot and earns zero.
- Insert images directly into the Word file, sized so the text is readable at 100% zoom. Do not attach them as separate files.
- Label each image with its code (SS-01, SS-02, ...) as a caption underneath.

Required screenshot list

Code	What it must show	Part
SS-01	VirtualBox Settings → Storage with both new disks attached to the SATA controller	A
SS-02	Virtual Media Manager listing storage1.vdi (20 GB) and storage2.vdi (30 GB)	A
SS-03	Output of <code>lsblk</code> showing the two new empty disks	B
SS-04	Output of <code>sudo fdisk -l</code> for the new disks only	B
SS-05	The <code>fdisk</code> session where you write the partition table on Storage 1	C
SS-06	Output of <code>sudo mkfs.ext4</code> on Storage 1's partition	C
SS-07	The <code>fdisk</code> session where you create both partitions on Storage 2	D
SS-08	The <code>fdisk p</code> command showing both partitions and their sizes	D
SS-09	Output of <code>mkfs.ext4</code> and <code>mkfs.ntfs</code> on Storage 2's two partitions	D
SS-10	The three mount point directories you created, shown with <code>ls -l /mnt</code>	E
SS-11	Output of <code>lsblk -f</code> with all three partitions mounted and their file system types visible	E
SS-12	Output of <code>df -h</code> showing all three mounted partitions	E
SS-13	The contents of your three text files, displayed with <code>cat</code>	F
SS-14	Output of <code>lsblk</code> after unmounting, with the MOUNTPOINTS column empty for all three	G

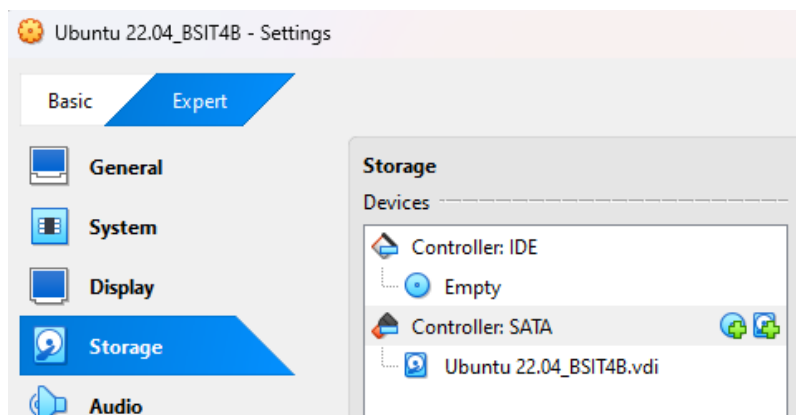
Part A — Attach the Two Virtual Disks

Time: 15 minutes | Evidence: SS-01, SS-02

A virtual disk in VirtualBox is a file on your host machine (.vdi) that the guest treats as if it were real hardware. Creating one is the equivalent of screwing a physical drive into a server chassis — the operating system will not do anything with it until you partition and format it.

Steps

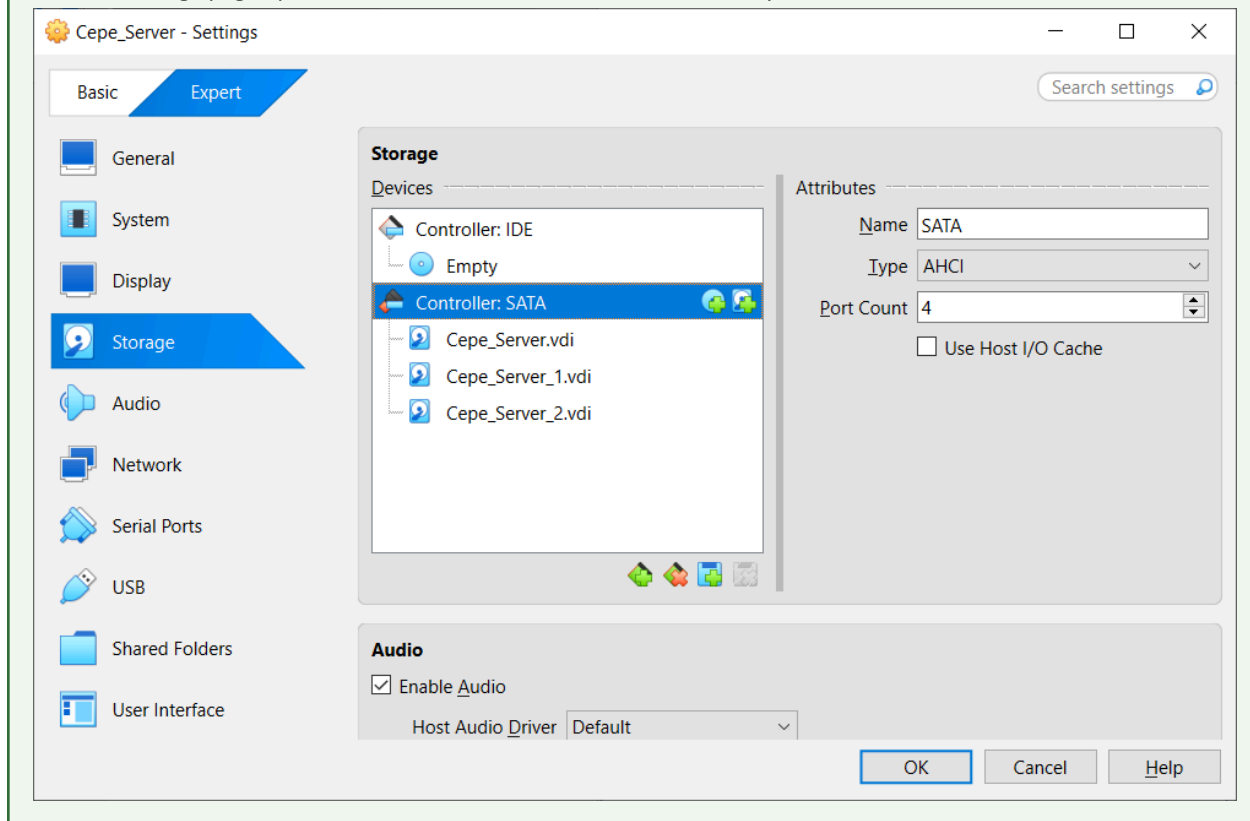
1. Make sure the VM is completely powered off. Not saved state — powered off.
2. Select the VM in VirtualBox Manager and click Settings, then Storage.
3. Click on Controller: SATA, then click the small hard-disk icon with the green plus sign (Add Hard Disk).
4. Choose Create, keep VDI as the type, keep Dynamically allocated, and set the size to 20 GB. Name the file storage1.
5. Repeat steps 3 and 4 to add a second disk of 30 GB. Name the file storage2.
6. Confirm the Storage tree now lists three disks under Controller: SATA — your original Ubuntu disk plus storage1.vdi and storage2.vdi.



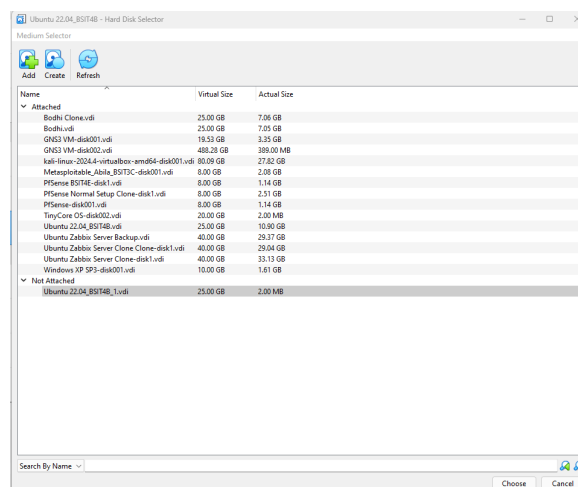
The Storage page of the VM settings. New disks are added to the SATA controller.

CAPTURE SS-01 NOW

With the Storage page open and both new disks visible in the tree, take your screenshot.



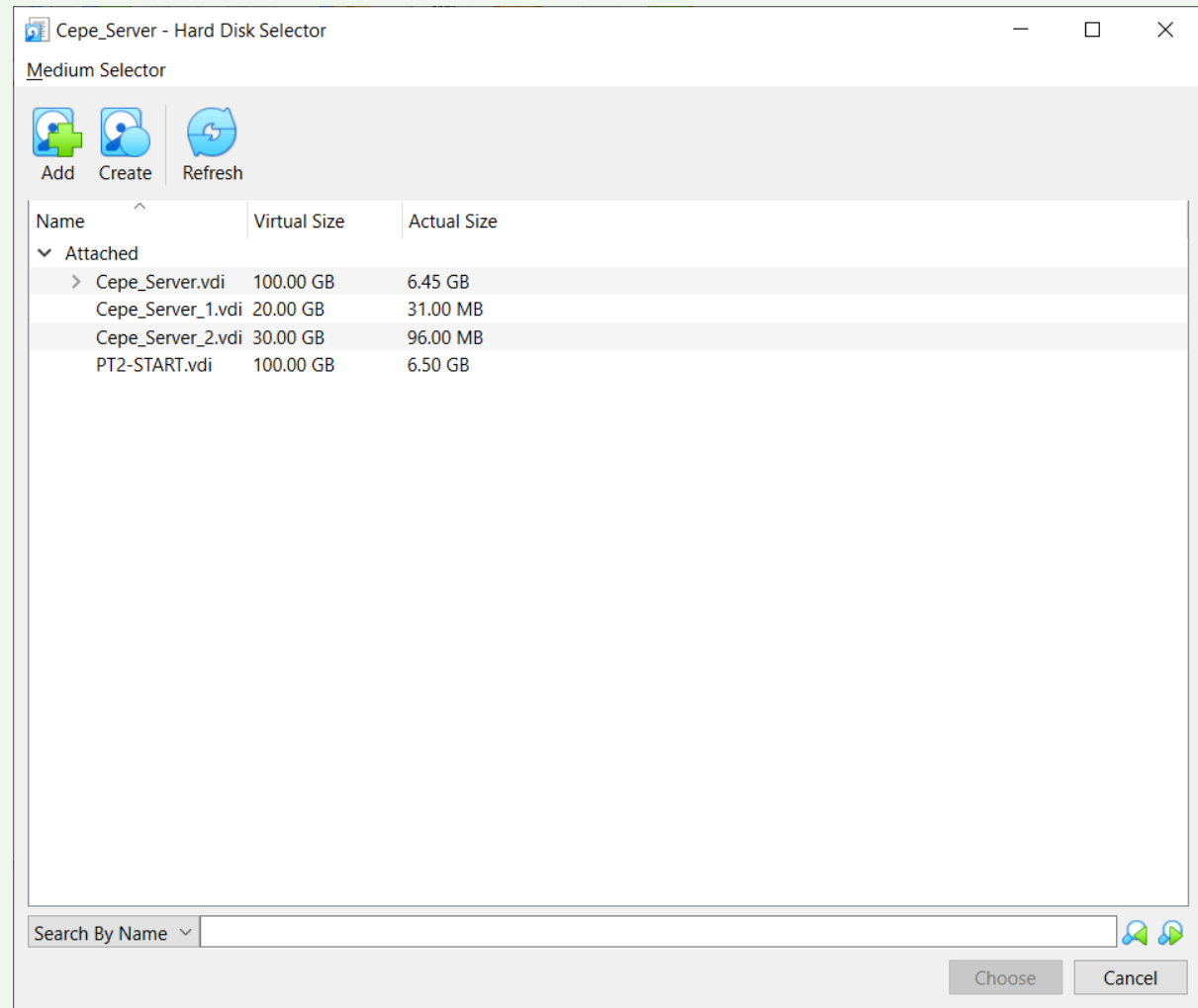
Now open File → Virtual Media Manager from the main VirtualBox window. Both new .vdi files should be listed with their virtual sizes.



The Virtual Media Manager lists every virtual disk known to VirtualBox.

CAPTURE SS-02 NOW

Screenshot the Virtual Media Manager showing storage1.vdi at 20 GB and storage2.vdi at 30 GB.



Record what you did

Item	Your answer
Full path of storage1.vdi on the host	Cepe_Server_1.vdi (located in your VirtualBox VM folder, e.g., C:\Users\USER\VirtualBox VMs\Cepe_Server\Cepe_Server_1.vdi)
Full path of storage2.vdi on the host	Cepe_Server_2.vdi (located in your VirtualBox VM folder, e.g., C:\Users\USER\VirtualBox VMs\Cepe_Server\Cepe_Server_2.vdi)
Allocation type chosen (dynamic / fixed)	Dynamically allocated
Controller the disks were attached to	SATA (or SATA Controller)

Part B — Boot Up and Identify the New Disks

Time: 10 minutes | Evidence: SS-03, SS-04

Start the VM and log in. The disks are attached, but Linux has not been told what to do with them yet — they exist as device files under `/dev` and nothing more.

```
ubuntu-server: ~
whoami
lsblk
sudo fdisk -l | less
```

In the `lsblk` output you are looking for two devices with the right sizes and, importantly, with nothing listed underneath them. A disk with no child entries has no partition table.

```
ian@ian-VirtualBox:/dev$ ls sd*
sda  sda1  sda2  sda3
```

Disks and partitions appear as files under `/dev` — `sda` is the first drive, `sda1` is its first partition.

READ THE NAMES CAREFULLY

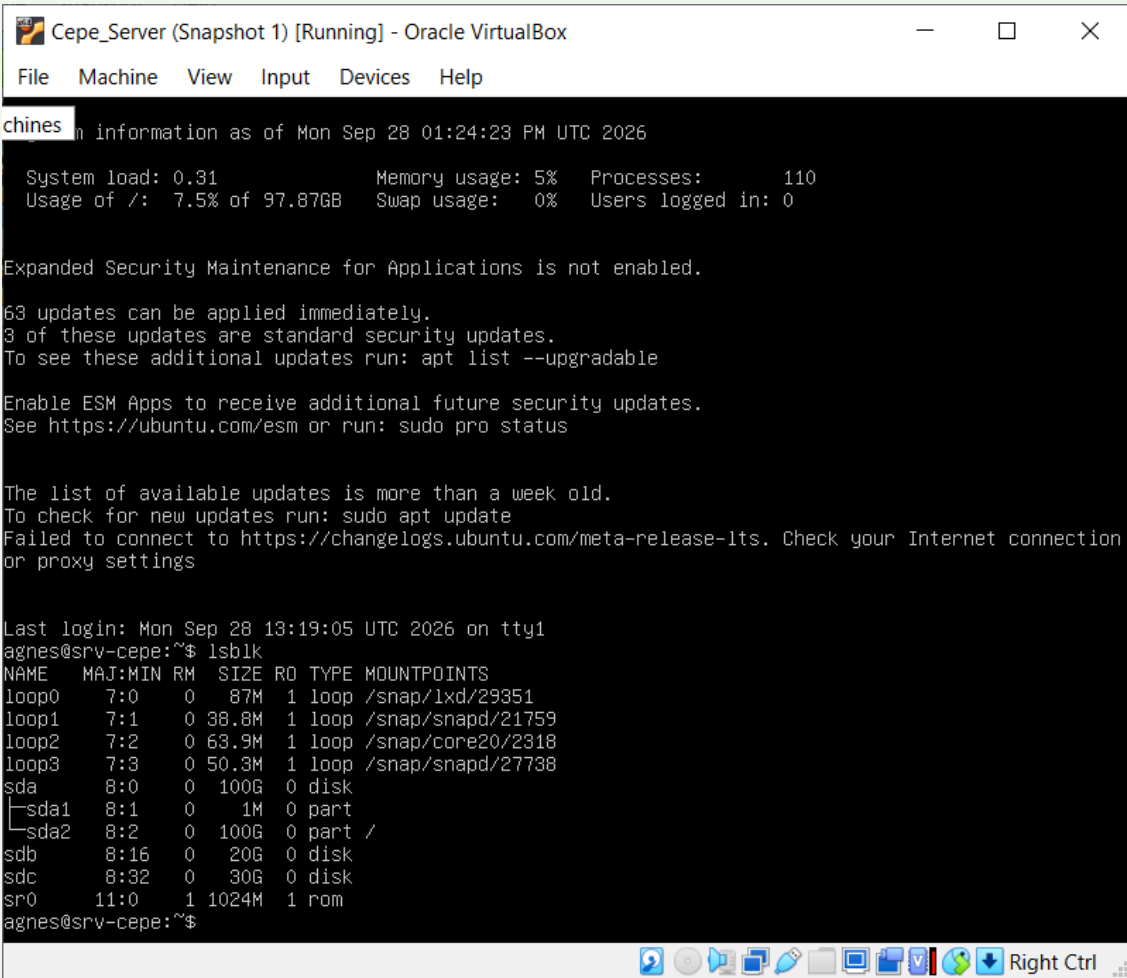
On almost every setup your existing system disk is `/dev/sda`, the 20 GB disk is `/dev/sdb` and the 30 GB disk is `/dev/sdc`. Do not assume this. Match by SIZE in the `lsblk` output, then write the confirmed names into the table below and use your own names for the rest of the task.

Confirm your device names

Disk	Size shown by lsblk	Device name on YOUR machine	Partitions listed?
Existing system disk	100G	/dev/sda	Yes (sda1, sda2)
Storage 1 (20 GB)	20G	/dev/sdb	No
Storage 2 (30 GB)	30G	/dev/sdc	No

CAPTURE SS-03 AND SS-04 NOW

SS-03: the full lsblk output.



```
chines@srv-cepe:~$ cat /etc/os-release
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=22.04
DISTRIB_CODENAME=jammy
DISTRIB_DESCRIPTION="Ubuntu 22.04.2 LTS"

System load: 0.31          Memory usage: 5%    Processes:      110
Usage of /:  7.5% of 97.87GB Swap usage:   0%    Users logged in: 0

Expanded Security Maintenance for Applications is not enabled.

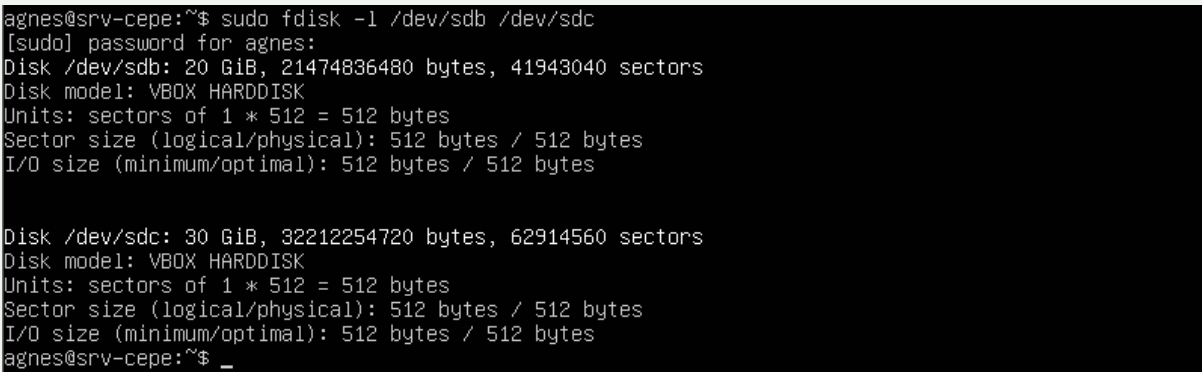
63 updates can be applied immediately.
3 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection
or proxy settings

Last login: Mon Sep 28 13:19:05 UTC 2026 on tty1
agnes@srv-cepe:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0   87M  1 loop /snap/lxd/29351
loop1       7:1    0  38.8M  1 loop /snap/snapd/21759
loop2       7:2    0  63.9M  1 loop /snap/core20/2318
loop3       7:3    0  50.3M  1 loop /snap/snapd/27738
sda          8:0    0 100G  0 disk
├─sda1       8:1    0    1M  0 part
└─sda2       8:2    0 100G  0 part /
sdb          8:16   0   20G  0 disk
sdc          8:32   0   30G  0 disk
sr0         11:0    1 1024M  1 rom
agnes@srv-cepe:~$
```

SS-04: the fdisk -l section for the two new disks, showing the message that they do not contain a recognised partition table.



```
agnes@srv-cepe:~$ sudo fdisk -l /dev/sdb /dev/sdc
[sudo] password for agnes:
Disk /dev/sdb: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdc: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
agnes@srv-cepe:~$ _
```

Part C — Storage 1: One Partition, Formatted ext4

Time: 20 minutes | Evidence: SS-05, SS-06

Storage 1 gets the simplest possible layout: one partition using the whole 20 GB, formatted ext4. This becomes the company archive.

C.1 Create the partition table and the partition

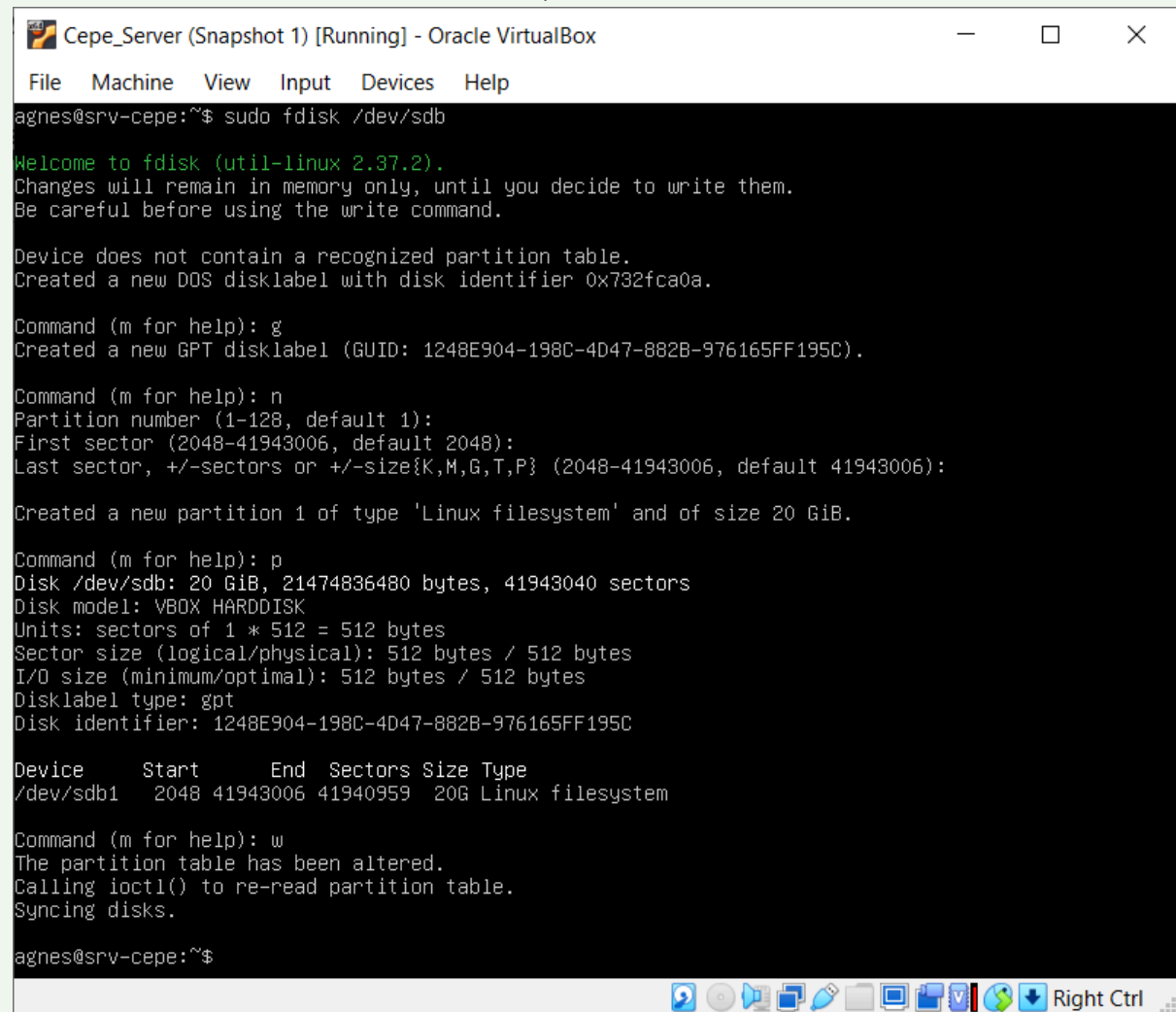
```
ubuntu-server: ~  
# use your own device name if it differs from sdb  
sudo fdisk /dev/sdb
```

Inside fdisk, type these one at a time, pressing Enter after each:

Type this	What it does
g	Create a new empty GPT partition table (this wipes the disk)
n	Add a new partition
(Enter) three times	Accept partition number 1, the default first sector, and the default last sector — which uses the entire disk
p	Print the table so you can check your work before committing
w	Write the table to disk and exit
q	Quit WITHOUT saving — use this whenever you make a mistake

CAPTURE SS-05 NOW

Screenshot the fdisk session — the commands you typed plus the confirmation that the partition table has been written. Then run `lsblk /dev/sdb` to confirm the new partition exists.



```
agnes@srv-cepe:~$ sudo fdisk /dev/sdb

Welcome to fdisk (util-linux 2.37.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x732fca0a.

Command (m for help): g
Created a new GPT disklabel (GUID: 1248E904-198C-4D47-882B-976165FF195C).

Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-41943006, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943006, default 41943006):

Created a new partition 1 of type 'Linux filesystem' and of size 20 GiB.

Command (m for help): p
Disk /dev/sdb: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 1248E904-198C-4D47-882B-976165FF195C

Device      Start      End  Sectors  Size Type
/dev/sdb1   2048 41943006 41940959   20G Linux filesystem

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

agnes@srv-cepe:~$
```

C.2 Format the partition as ext4

A partition is only a reserved region of the disk. Formatting installs a file system into that region. Until you format it, the partition cannot hold a single file.

```
ubuntu-server: ~
sudo mkfs.ext4 /dev/sdb1

# then confirm the file system type is now visible:
lsblk -f /dev/sdb
```

Partition device name	Size reported by lsblk	First 8 characters of the UUID
sdb1 (/dev/sdb1)	20G	99dc1d32

CAPTURE SS-06 NOW

The mkfs.ext4 output plus the lsblk -f result showing ext4 in the FSTYPE column.

```
agnes@srv-cepe:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 5242619 4k blocks and 1310720 inodes
Filesystem UUID: 99dc1d32-ff05-4eae-ad64-f172e397ff08
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000

Allocating group tables: done
Writing inode tables: done
Creating journal (32768 blocks): done
Writing superblocks and filesystem accounting information: done

agnes@srv-cepe:~$ lsblk -f /dev/sdb
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sdb
└─sdb1 ext4   1.0           99dc1d32-ff05-4eae-ad64-f172e397ff08
agnes@srv-cepe:~$ _
```

Part D — Storage 2: Two Partitions, Two File Systems

Time: 35 minutes | Evidence: SS-07, SS-08, SS-09

This is the part that separates a technician from someone who can only follow a wizard. One physical disk, split into two logical regions, each carrying a different file system, each serving a different purpose.

D.1 Install NTFS support

Ubuntu Server does not ship with the tool that creates NTFS file systems. Install it now, before you need it.

```
ubuntu-server: ~  
sudo apt update  
sudo apt install -y ntfs-3g  
  
# verify the formatting tool is present:  
which mkfs.ntfs
```

D.2 Create both partitions

Start fdisk against **/dev/sdc** — again, use your own confirmed device name.

```
ubuntu-server: ~  
sudo fdisk /dev/sdc
```

Type this	What it does
g	New empty GPT partition table
n	Add the first partition
(Enter)	Accept partition number 1
(Enter)	Accept the default first sector
+15G	Make this partition exactly 15 GB
n	Add the second partition
(Enter)	Accept partition number 2
(Enter)	Accept the default first sector
(Enter)	Accept the default last sector — takes all remaining space
t	Change a partition type
2	Apply the change to partition 2
L	List the available type codes and find Microsoft basic data
(its number)	Enter the number shown next to Microsoft basic data — usually 11
p	Print the table and check both sizes before committing
w	Write and exit

WHY CHANGE THE PARTITION TYPE?

The type code is a label stored in the partition table. It does not format anything and it does not stop Linux from using the partition. What it does is tell other operating systems what to expect. Set it correctly and a Windows machine recognises the partition immediately; leave it as Linux filesystem and Windows may offer to format it instead.

YOUR SECOND PARTITION WILL NOT BE EXACTLY 15 GB

Expect something close to 14.9 GB. That is correct behaviour, not an error, and Concept Question 3 asks you to explain it. Do not go back and try to force the numbers to match.

D.3 Record the layout before you format

Partition	Device name	Size from fdisk p	Type code shown
Storage 2, partition 1	/dev/sdc1	15G	Linux filesystem
Storage 2, partition 2	/dev/sdc2	15G	Microsoft basic data

CAPTURE SS-07 AND SS-08 NOW

SS-07: the fdisk session with both partitions being created.

```
agnes@srv-cepe:~$ sudo fdisk /dev/sdc

Welcome to fdisk (util-linux 2.37.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): g
Created a new GPT disklabel (GUID: A3CA7E44-7C4C-C04D-BADA-D7C67D04B3C6).


Command (m for help): n
Partition number (1-128, default 1):
First sector (2048-62914526, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-62914526, default 62914526): +15G

Created a new partition 1 of type 'Linux filesystem' and of size 15 GiB.


Command (m for help): n
Partition number (2-128, default 2):
First sector (31459328-62914526, default 31459328):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-62914526, default 62914526):

Created a new partition 2 of type 'Linux filesystem' and of size 15 GiB.


Command (m for help):
```

SS-08: the output of the `p` command showing both partitions, their sizes and their type codes.

```
Command (m for help): t
Partition number (1,2, default 2): 2
Partition type or alias (type L to list all): 11

Changed type of partition 'Linux filesystem' to 'Microsoft basic data'.


Command (m for help): p
Disk /dev/sdc: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: A3CA7E44-7C4C-C04D-BADA-D7C67D04B3C6


Device            Start       End   Sectors  Size Type
/dev/sdc1         2048 31459327 31457280  15G Linux filesystem
/dev/sdc2      31459328 62914526 31455199  15G Microsoft basic data


Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

agnes@srv-cepe:~$
```

D.4 Format each partition with its own file system

```
ubuntu-server: ~
# Partition 1 -> ext4, for Finance records
sudo mkfs.ext4 /dev/sdc1

# Partition 2 -> NTFS, for the Windows exchange folder
sudo mkfs.ntfs -f -L IVT_EXCHANGE /dev/sdc2

# confirm both:
lsblk -f
```

THE -f FLAG IS IMPORTANT

Without **-f**, mkfs.ntfs performs a full format and zeroes every sector on the partition. On 15 GB that can take twenty minutes or more, and you do not have twenty minutes. The **-f** flag performs a quick format instead, which is what you want here. **-L IVT_EXCHANGE** sets a volume label so the partition is easy to recognise later.

CAPTURE SS-09 NOW

One screenshot containing both format commands and the lsblk -f output that shows ext4 on partition 1 and ntfs on partition 2.

```
agnes@srv-cepe:~$ sudo mkfs.ext4 /dev/sdc1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: 479cd3e3-3380-4ac3-a2c6-a56bee12ce3e
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

agnes@srv-cepe:~$ sudo mkfs.ntfs -f /dev/sdc2
Cluster size has been automatically set to 4096 bytes.
Creating NTFS volume structures.
mkntfs completed successfully. Have a nice day.
agnes@srv-cepe:~$ _
```

Part E — Create Mount Points and Mount Everything

Time: 20 minutes | Evidence: SS-10, SS-11, SS-12

A formatted partition is still unreachable. Linux has no drive letters — there is no D: to click on. Instead you attach the partition to an empty directory somewhere in the single tree that starts at /. That directory is the mount point, and after mounting, everything you write inside it lands on the partition rather than on the system disk.

E.1 Decide and create your mount points

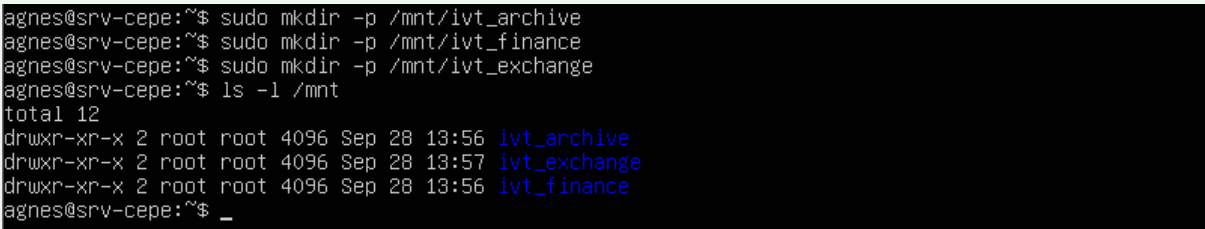
Use **/mnt** — the directory that exists specifically for storage an administrator mounts by hand. (**/media** is for removable devices the system mounts automatically.) Name each directory after the department it serves.

```
ubuntu-server: ~
sudo mkdir -p /mnt/ivt_archive
sudo mkdir -p /mnt/ivt_finance
sudo mkdir -p /mnt/ivt_exchange

ls -l /mnt
```

CAPTURE SS-10 NOW

Screenshot the ls -l /mnt output showing all three empty directories.



```
agnes@srv-cepe:~$ sudo mkdir -p /mnt/ivt_archive
agnes@srv-cepe:~$ sudo mkdir -p /mnt/ivt_finance
agnes@srv-cepe:~$ sudo mkdir -p /mnt/ivt_exchange
agnes@srv-cepe:~$ ls -l /mnt
total 12
drwxr-xr-x 2 root root 4096 Sep 28 13:56 ivt_archive
drwxr-xr-x 2 root root 4096 Sep 28 13:57 ivt_exchange
drwxr-xr-x 2 root root 4096 Sep 28 13:56 ivt_finance
agnes@srv-cepe:~$ _
```

E.2 Mount all three partitions

```
ubuntu-server: ~
sudo mount /dev/sdb1 /mnt/ivt_archive
sudo mount /dev/sdc1 /mnt/ivt_finance
sudo mount /dev/sdc2 /mnt/ivt_exchange

lsblk -f
df -h
```

The mount command produces no output when it succeeds. Silence means it worked. If you get an error, read it before doing anything else — most mount failures name the exact problem.

E.3 Fill in your mount map

Partition	Device	Mount point	File system	Size in df -h	✓
Company archive	/dev/sdb1	/mnt/ivt_archive	ext4	20G	/
Finance records	/dev/sdc1	/mnt/ivt_finance	ext4	15G	/

Partition	Device	Mount point	File system	Size in df -h	✓
Windows exchange	/dev/sdc2	/mnt/ivt_exchange	ntfs	15G	/

CAPTURE SS-11 AND SS-12 NOW

SS-11: lsblk -f with all three file system types and mount points. SS-12: df -h with all three mounted.

```

agnes@srv-cepe:~$ sudo mount /dev/sdb1 /mnt/ivt_archive
mount: /mnt/ivt_archive: /dev/sdb1 already mounted on /mnt/ivt_archive.
agnes@srv-cepe:~$ sudo mount /dev/sdc1 /mnt/ivt_finance
agnes@srv-cepe:~$ sudo mount /dev/sdc2 /mnt/ivt_exchange
agnes@srv-cepe:~$ lsblk -f
NAME        FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0       squashfs 4.0                                     0      100% /snap/lxd/29351
loop1       squashfs 4.0                                     0      100% /snap/snapd/21759
loop2       squashfs 4.0                                     0      100% /snap/core20/2318
loop3       squashfs 4.0                                     0      100% /snap/snapd/27738
sda
├─sda1
└─sda2 ext4      1.0                be95dfd1-963c-4bf5-9e93-0d2b71362316    85.5G      8% /
sdb
└─sdb1 ext4      1.0                99dc1d32-ff05-4eae-ad64-f172e397ff08    18.5G      0% /mnt/ivt_archive
sdc
├─sdc1 ext4      1.0                479cd3e3-3380-4ac3-a2c6-a56bee12ce3e    13.9G      0% /mnt/ivt_finance
└─sdc2 ntfs      217B4FE05B83E959    14.9G      0% /mnt/ivt_exchange
sr0
agnes@srv-cepe:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            392M  1.1M  391M   1% /run
/dev/sda2        98G   7.4G   86G   8% /
tmpfs            2.0G     0   2.0G   0% /dev/shm
tmpfs            5.0M     0   5.0M   0% /run/lock
tmpfs            392M  4.0K  392M   1% /run/user/1000
/dev/sdb1        20G   24K   19G   1% /mnt/ivt_archive
/dev/sdc1        15G   24K   14G   1% /mnt/ivt_finance
/dev/sdc2        15G   65M   15G   1% /mnt/ivt_exchange
agnes@srv-cepe:~$ _

```

Part F — Write One File to Each Partition

Time: 15 minutes | Evidence: SS-13

Mounting proves the partition is reachable. Writing a file proves it is usable. Create one text file on each of the three partitions, each containing your own identifying details so the file is unmistakably yours.

F.1 Create the three files

Use **cat** > to type content directly into a new file. Press **Ctrl+D** on a line of its own when you are finished — that signals end of input.

```
ubuntu-server: ~
sudo cat > /mnt/ivt_archive/archive_note.txt
Isla Verde Trading Corp - Company Archive
Prepared by: <your full name>
Section: <your section>      Date: <today>
Device: /dev/sdb1   File system: ext4
[press Ctrl+D here]
```

Repeat for the other two partitions, changing the device, file system and folder each time:

```
ubuntu-server: ~
sudo cat > /mnt/ivt_finance/finance_note.txt
sudo cat > /mnt/ivt_exchange/exchange_note.txt
```

F.2 Verify all three files

```
ubuntu-server: ~
ls -l /mnt/ivt_archive /mnt/ivt_finance /mnt/ivt_exchange

cat /mnt/ivt_archive/archive_note.txt
cat /mnt/ivt_finance/finance_note.txt
cat /mnt/ivt_exchange/exchange_note.txt
```

IF PERMISSION IS DENIED

A freshly formatted ext4 partition is owned by root, so an ordinary user cannot write into it. You can either keep using **sudo** as shown above, or hand the directory to yourself with **sudo chown \$USER:\$USER /mnt/ivt_archive**. Ownership and permissions are the subject of Weeks 7 and 8 — for today, sudo is enough.

CAPTURE SS-13 NOW

One screenshot showing the contents of all three files displayed with cat, with your name visible in each.

```
agnes@srv-cepe:~$ cat > /mnt/ivt_archive/archive_note.txt
Isla Verde Trading Corp - Company Archive
Prepared by: Agnes
Section: BSIT
Date: Sept 28 2026
Device: /dev/sdb1 File system: ext4
agnes@srv-cepe:~$ _
```

```
agnes@srv-cepe:~$ cat > /mnt/ivt_finance/finance_note.txt
Isla Verde Trading Corp - Finance Records
Prepared by: Agnes
Section: BSIT
Date: Sep 28 2026
Device: /dev/sdc1 File system: ext4
```

```
agnes@srv-cepe:~$ cat > /mnt/ivt_finance/finance_note.txt
Isla Verde Trading Corp - Finance Records
Prepared by: Agnes
Section: BSIT
Date: Sep 28 2026
Device: /dev/sdc1 File system: ext4
agnes@srv-cepe:~$ cat > /mnt/ivt_exchange/exchange_note.txt
Isla Verde Trading Corp - Windows Exchange
Prepared by: Agnes
Section: BSIT
Date: Sept 28 2026
Device: /dev/sdc2 File system: ntfs
agnes@srv-cepe:~$
```

```
agnes@srv-cepe:~$ ls -l /mnt/ivt_archive /mnt/ivt_finance /mnt/ivt_exchange
/mnt/ivt_archive:
total 20
-rw-r--r-- 1 agnes agnes 130 Sep 28 14:14 archive_note.txt
drwx----- 2 agnes agnes 16384 Sep 28 13:32 lost+found
-rw-r--r-- 1 agnes agnes 0 Sep 28 14:04 test.txt

/mnt/ivt_exchange:
total 1
-rwxrwxrwx 1 root root 131 Sep 28 14:23 exchange_note.txt
-rwxrwxrwx 1 root root 0 Sep 28 14:05 test.txt

/mnt/ivt_finance:
total 20
-rw-rw-r-- 1 agnes agnes 129 Sep 28 14:21 finance_note.txt
drwx----- 2 agnes agnes 16384 Sep 28 13:53 lost+found
-rw-r--r-- 1 agnes agnes 0 Sep 28 14:04 test.txt
agnes@srv-cepe:~$ cat /mnt/ivt_archive/archive_note.txt
Isla Verde Trading Corp - Company Archive
Prepared by: Agnes
Section: BSIT
Date: Sept 28 2026
Device: /dev/sdb1 File system: ext4
agnes@srv-cepe:~$ cat /mnt/ivt_finance/finance_note.txt
Isla Verde Trading Corp - Finance Records
Prepared by: Agnes
Section: BSIT
Date: Sep 28 2026
Device: /dev/sdc1 File system: ext4
```

```
agnes@srv-cepe:~$ cat /mnt/ivt_exchange/exchange_note.txt
Isla Verde Trading Corp - Windows Exchange
Prepared by: Agnes
Section: BSIT
Date: Sept 28 2026
Device: /dev/sdc2 File system: ntfs
```

Part G — Unmount Everything Cleanly

Time: 10 minutes | Evidence: SS-14

When you write a file, Linux does not necessarily push it to the disk immediately. It holds data in memory and writes it out when convenient, because that is dramatically faster. Unmounting forces every pending write to complete and then detaches the file system. This is why Linda's thesis went missing — the USB was pulled before the write finished.

```
ubuntu-server: ~
# make sure your shell is NOT sitting inside any mount point:
cd /

sudo umount /mnt/ivt_archive
sudo umount /mnt/ivt_finance
sudo umount /mnt/ivt_exchange

lsblk
df -h
```

"TARGET IS BUSY"

Something is still using the file system — usually your own shell sitting in that directory. Run `cd /` and try again. To find the culprit, use `lssof +D /mnt/ivt_archive`. Never force an unmount just to make the message go away — that is how data gets corrupted.

Final verification

Check	Expected result	✓
lsblk MOUNTPOINTS for sdb1, sdc1, sdc2	All three empty	/
df -h	None of the three partitions appear	/
ls /mnt/ivt_archive	Empty — the directory still exists but the file is not visible	/
FSTYPE column in lsblk -f	Still shows ext4 / ext4 / ntfs	/

THE LAST CHECK IS THE MOST IMPORTANT ONE

After unmounting, the mount point directory is empty — but your file has not been deleted. Understanding why is the whole point of this task, and Concept Question 7 asks you to explain it.

CAPTURE SS-14 NOW

Screenshot lsblk after unmounting, with the MOUNTPOINTS column empty for all three partitions.

```
agnes@srv-cepe:~$ sudo umount /mnt/ivt_archive
agnes@srv-cepe:~$ lsblk -f
NAME        FSTYPE     FSVER LABEL  UUID                                  FSAVAIL  FSUSE% MOUNTPOINTS
loop0       squashfs   4.0                                0        100% /snap/snapd/21759
loop1       squashfs   4.0                                0        100% /snap/core20/2318
loop2       squashfs   4.0                                0        100% /snap/snapd/27738
loop3       squashfs   4.0                                0        100% /snap/lxd/29351
sda
├─sda1
├─sda2 ext4      1.0      be95dfd1-963c-4bf5-9e93-0d2b71362316 85.5G    8% /
├─sdb
├─sdb1 ext4      1.0      99dc1d32-ff05-4eae-ad64-f172e397ff08
├─sdc
├─sdc1 ext4      1.0      479cd3e3-3380-4ac3-a2c6-a56bee12ce3e
├─sdc2 ntfs
└─sdc2 217B4FE05B83E959
sr0
agnes@srv-cepe:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            392M  1.1M  391M   1% /run
/dev/sda2        98G   7.4G   86G   8% /
tmpfs            2.0G    0    2.0G   0% /dev/shm
tmpfs            5.0M    0    5.0M   0% /run/lock
tmpfs            392M  4.0K  392M   1% /run/user/1000
```

ubuntu-server: ~

finally, shut the VM down properly, then take a snapshot named PT3-DONE

sudo shutdown -h now

Part H — Concept Questions

Time: 15 minutes | 30 of the 100 points

Answer in your own words, in complete sentences, in the space provided under each question. Three to five sentences is the right length — enough to show reasoning, short enough to stay focused. Answers copied from the slides or from a classmate earn zero for the entire section.

- Q1. Linux gave your new disks the names sdb and sdc, and the partitions on the second one became sdc1 and sdc2. Break down the name /dev/sdc2 piece by piece and explain what each part tells you. Then explain why Linux uses this scheme instead of the drive letters D: and E: that Windows would have assigned.
- **In /dev/sdc2, /dev/ is the location for device files, sd means a SATA/SCSI storage drive, c stands for the third hard drive connected to the computer, and 2 means the second partition on that drive. Linux uses this scheme because it mounts all partitions into a single root file system (/) instead of using separate drive letters like Windows (D:, E:). This keeps all storage structured in one place no matter how many drives you add.**
-
- Q2. You created a partition, and then you formatted it. Explain the difference between these two operations. Include in your answer what would happen if you tried to mount a partition that had been created but never formatted.
- **Partitioning divides a physical disk into logical sections, while formatting creates a file system (like ext4 or NTFS) inside a partition so the OS can save and read files. If you try to mount an unformatted partition, Linux will reject it because there is no readable file system structure. The system will throw an error such as "unknown filesystem type" and refuse to open it.**
-
- Q3. Storage 2 is a 30 GB disk and you asked for a 15 GB first partition, yet the second partition came out slightly under 15 GB. Explain where the missing space went. Your answer should mention at least two things that consume space on a disk before any user file is written.
- **The missing disk space was taken up by file system overhead and partition alignment rules before any files were saved. Formatting automatically reserves space for system metadata like inode tables, superblocks, and journaling files. Additionally, drive makers measure size in decimal units while Linux calculates in binary units, making the drive show a slightly smaller available capacity.**
-
- Q4. Your instructor asks why the exchange partition uses NTFS when everything else on the server is ext4. Give the argument in favour of the NTFS choice for that specific partition, then give one real disadvantage the Isla Verde team accepts by making it. Be specific about what ext4 provides that NTFS-on-Linux does not.
- **NTFS is chosen for the exchange partition because it provides cross-platform compatibility, allowing Windows systems to easily read and write files on that shared drive without extra drivers. The main disadvantage for Isla Verde is that Linux handles NTFS through FUSE user-space drivers, which results in slower file transfers and weaker native permission management. Native ext4 provides deep Linux system integration, better disk performance, and built-in POSIX file permissions that NTFS on Linux cannot fully support.**
-
- Q5. A colleague suggests you skip creating /mnt/ivt_finance and instead mount the Finance partition directly onto /home, since that directory already exists and already has files in it. Explain what would happen to the existing contents of /home, and why mount points should normally be empty directories.
- **If you mount the Finance partition directly onto /home, the existing files inside /home will be temporarily hidden by the newly attached file system. The old files remain safe on disk, but users will lose access to their personal home directories until the partition is unmounted. Mount points should normally be empty directories so that no existing files are hidden or rendered inaccessible when a partition is attached.**
-

Q6. Linda saved her thesis in Ubuntu, could not find it in Windows, and later got a "No such file or directory" error when the filename looked correct to her. Identify the two separate technical causes behind these two symptoms, and state which of them your NTFS partition is designed to prevent.

- **The first cause is that Windows cannot natively read Linux ext4 partitions, hiding files saved on an ext4 drive when booted into Windows. The second cause is that Linux is case-sensitive (Thesis.docx and thesis.docx are different files), while Windows is case-insensitive. Your NTFS partition is specifically designed to solve the first issue by providing a shared filesystem that both Windows and Linux can read and write to.**
-

Q7. After you unmounted everything in Part G, the command `ls /mnt/ivt_archive` returned nothing — yet the text file you created has not been deleted. Explain where the file actually is and what mounting does. Then state what you would need to type to see the file again.

- **The text file is physically stored inside the filesystem on the underlying hard drive block device (`/dev/sdb1`), not in the system folder itself. Mounting bridges a storage partition to an empty directory in Linux so the operating system can read and write its contents. To see the file again, you need to remount the partition by typing `sudo mount /dev/sdb1 /mnt/ivt_archive` and then running `cat /mnt/ivt_archive/archive_note.txt`.**
-

Q8. Vince, the intern from the case study, was told the server was out of space. He guessed which directory was the problem and deleted files from `/bin`, which broke the server completely. Name the single command he should have run first, state exactly what information it would have given him, and explain how that information would have changed his decision.

- **Vince should have run the `df -h` command first. This command displays disk space usage across all mounted filesystems in a human-readable format, showing exact percentage levels for each partition. This information would have shown him that the root directory (`/`) or `/bin` wasn't full and pointed him directly to the actual full partition (like `/var` or `/home`), preventing him from deleting critical binary files in `/bin` and breaking the server.**
-

Q9. You try to unmount a partition and get the message "target is busy". Explain what that message means, describe the correct procedure for resolving it, and explain the specific risk of using a forced unmount to bypass it instead.

- **The "target is busy" message means a process, user session, or open terminal is currently accessing files or working inside that mounted directory, preventing Linux from safely detaching it. The correct procedure to resolve this is to identify and stop the active process using `fuser -m /mnt/mountpoint` or `lsof /mnt/mountpoint`, or simply change out of the directory in your terminal using `cd /`. Using a forced unmount (`umount -f` or `-l`) carries the risk of corrupting open files, losing unwritten data stored in RAM buffer caches, or crashing applications relying on that filesystem.**
-

Q10. Reboot the VM and check `/mnt` with `lsblk`. Your three partitions are no longer mounted, even though you never touched them. Explain why mounts do not survive a reboot, and name the configuration file an administrator would edit to make them permanent.

- **Manual mounts do not survive a reboot because the mount command only attaches filesystems in temporary system memory (RAM), which is completely cleared when the computer powers off or restarts. To make mounts permanent across system reboots, a system administrator must edit the `/etc/fstab` configuration file. Adding the partition details (UUID, mount point, filesystem type, and mount options) to this file tells Linux to automatically mount the partitions every time the operating system boots up.**
-

5 What to Submit

One Microsoft Word document (.docx). Nothing else — no zip archives, no loose image files, no PDFs.

File name

Naming convention

PT3_FileSystem_<LASTNAME>_<FIRSTNAME>_<SECTION>.docx

example:

PT3_FileSystem_DELACRUZ_JUAN_BSIT4A.docx

Required contents, in this order

#	Section	Must contain
1	Cover page	Your name, section, date performed, date submitted, title of the task
2	Part A	SS-01, SS-02, and the completed disk record table
3	Part B	SS-03, SS-04, and the completed device name table
4	Part C	SS-05, SS-06, and the completed Storage 1 table
5	Part D	SS-07, SS-08, SS-09, and the completed layout table
6	Part E	SS-10, SS-11, SS-12, and the completed mount map
7	Part F	SS-13
8	Part G	SS-14 and the completed final verification checklist
9	Part H	All ten concept questions answered
10	Declaration	The signed statement below

Declaration

I performed this task myself on my own virtual machine. Every screenshot in this document was captured from my own screen during my own session. I understand that submitting another student's screenshots, or output that has been retyped rather than captured, is academic dishonesty and results in a grade of zero for the entire task.

AGNES JOY T. CEPE
Signature over printed name

October 1, 2026
Date

Deadline and late work

When submitted	Maximum score
On or before the deadline	100%
Up to 24 hours late	90%
24 to 72 hours late	75%
More than 72 hours late	Not accepted without an approved excuse

Appendix A — When Things Go Wrong

What you see	What it means	What to do
The new disks do not appear in lsblk after booting	The disks were attached while the VM was running, or attached to the wrong controller	Power the VM fully off, re-check Settings → Storage, boot again
fdisk: cannot open /dev/sdb: Permission denied	You forgot sudo	Re-run the command with sudo in front
Device or resource busy when running fdisk	A partition on that disk is currently mounted	Unmount it first, then run fdisk again
mkfs.ntfs: command not found	The ntfs-3g package is not installed	sudo apt update && sudo apt install -y ntfs-3g
mkfs.ntfs is taking forever	You omitted the -f flag, so it is zeroing every sector	Press Ctrl+C and re-run with -f
mount: /mnt/x: wrong fs type, bad option, bad superblock	The partition was never formatted, or you named the wrong device	Check lsblk -f — if FSTYPE is blank, format it first
mount: /mnt/x: mount point does not exist	You skipped the mkdir step or mistyped the path	Create the directory with sudo mkdir -p, then mount again
umount: target is busy	A process — probably your own shell — is inside the mount point	cd / then try again; use lsof +D <path> to identify the culprit
Permission denied when creating the text file	The freshly formatted partition is owned by root	Use sudo, or chown the directory to your user
Your partition sizes look wrong after a reboot	You are reading a different device — names can shift	Match by size and UUID with lsblk -f, not by name alone

Appendix B — Command Reference

Looking at storage

Command	What it does
lsblk	Tree of all block devices, their sizes and where they are mounted
lsblk -f	The same tree plus file system type, label and UUID
sudo fdisk -l	Detailed partition table for every disk
df -h	Space used and available on every mounted file system
mount	Every file system currently mounted, with its options
ls /dev/sd*	The raw device files for disks and partitions

Changing storage

Command	What it does
sudo fdisk /dev/sdX	Open the interactive partition editor for one disk
sudo mkfs.ext4 /dev/sdXn	Format a partition with the ext4 file system
sudo mkfs.ntfs -f /dev/sdXn	Quick-format a partition with NTFS
sudo mkdir -p /mnt/name	Create a mount point directory
sudo mount /dev/sdXn /mnt/name	Attach a partition to a mount point
sudo umount /mnt/name	Detach it and flush all pending writes

Inside fdisk

Key	Action	Key	Action
g	New GPT partition table	t	Change a partition's type code
o	New MS-DOS partition table	L	List available type codes
n	Create a new partition	p	Print the current table
d	Delete a partition	w	Write changes and exit
m	Show the help menu	q	Quit WITHOUT saving

SIZE SHORTHAND IN FDISK

When fdisk asks for the last sector you can type a size instead: **+15G** for 15 gibibytes, **+500M** for 500 mebibytes, or just press Enter to use everything that is left.

PERFORMANCE TASK 3 — SCORING RUBRIC

IT 4112 System Administration and Maintenance • File System Administration • 100 points

Criterion	Exemplary (100%)	Proficient (80%)	Developing (60%)	Beginning (40%)
Disk Provisioning & Identification 15 pts	Both disks created at exactly the correct sizes and attached to the SATA controller. Device names confirmed by size and recorded. SS-01 to SS-04 complete and readable.	Both disks attached at correct sizes. Device names identified with one minor recording error. All four screenshots present.	Disks attached but a size is wrong, or device names were assumed rather than verified. One screenshot missing.	Disks not attached correctly, or two or more screenshots missing. No evidence the OS detected the new hardware.
Partitioning Accuracy 25 pts	Storage 1 has one partition using the full disk. Storage 2 has two, the first exactly 15 GB, the second using all remaining space with the type code set to Microsoft basic data.	All partitions created with correct counts and near-correct sizes. Type code set. Minor gaps in the layout table.	Partitions created but one size is clearly wrong, or the second disk has the wrong partition count, or the type code was not changed.	Partitioning incomplete or done on the wrong device. Partition table not written, or evidence does not show the required layout.
Formatting & File Systems 20 pts	Storage 1 and Storage 2 partition 1 are ext4; Storage 2 partition 2 is NTFS with a volume label. lsblk -f confirms all three. The -f flag used correctly.	All three formatted with the correct file systems and confirmed by lsblk -f. Volume label missing or -f flag omitted.	Two of the three formatted correctly, or a file system type does not match the specification.	Formatting not performed, performed on the wrong partitions, or no evidence that any file system was created.
Mounting, Files & Clean Unmount 20 pts	Three purposefully named mount points under /mnt. All three mounted and verified in lsblk -f and df -h. An identifying text file on each, shown with cat. All unmounted cleanly and proven.	All three mounted, files written, and all unmounted. One verification step missing or one mount point poorly named.	Some partitions mounted or some files created, but the set is incomplete; or unmounting was not verified.	Mounting not achieved, files not written, or partitions left mounted with no clean-unmount evidence.
Concept Questions 15 pts	All ten answered in the student's own words with correct reasoning. Answers connect the commands performed to the concepts and to the case studies.	All ten answered. Technically sound overall with one or two shallow or partially incorrect explanations.	Seven to nine answered, or several answers restate the handout without showing understanding.	Fewer than seven answered, or answers show fundamental misunderstanding, or content is copied.
Documentation Quality 5 pts	Follows the required order exactly, all fourteen screenshots labelled, every record table completed, correct file name, declaration signed.	Well organised with all screenshots labelled. One or two tables incomplete, or a minor file-naming error.	Disorganised or several tables blank. Screenshots present but unlabelled or hard to read.	Required structure not followed, screenshots unlabelled or missing, declaration not signed.

PERFORMANCE TASK 3 — SCORE SHEET

To be completed by the instructor.

Criterion	Weight	Score	Comments
Disk Provisioning & Identification	15		
Partitioning Accuracy	25		
Formatting & File Systems	20		
Mounting, File Creation & Clean Unmount	20		
Concept Questions	15		
Documentation Quality	5		
TOTAL	100		

Student: _____ Section: _____ Date: _____

Evaluated by: _____ Signature: _____